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WEEK 9: Bluetooth Data Interfacing

SECTION 1

GROUP 11

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ABSTRACT

This experiment aims to investigate the use of Bluetooth data interfacing. The objective of this task is to create a wireless monitoring system that can be applied by using the communication of Bluetooth and ESP 32 and a smartphone. The sensor data can be read by potentiometer and will transmit it to the Bluetooth, and the sensor value is going to be shown in serial monitor of the device. The methodology consists of one task. In the task, the servo motor is being used so we can see how it can relate to the Bluetooth and the potentiometer works as sensor that will give us the sensor value on the serial monitor. Other than that, the serial Bluetooth terminal is being downloaded on the device so we can see the connections of ESP 32 and the potentiometer.

The key findings of this task are we learn how Bluetooth can incorporate into the IOT applications and how the servo and potentiometer can connect with Bluetooth. Other than that, the signal also has been classified into 3 segments which are low range (when the sensor value resets the servo motor to initial position) , Mid-range (when the sensor value makes the motor rotate 90 °) and High range when the motor keeps rotating 360 °.

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INTRODUCTION

The Internet of Things (IoT) is a technology that enables devices and electronic systems to communicate and exchange data through wireless communication networks. IoT systems are widely used in modern applications such as smart homes, industrial automation, healthcare systems, and remote monitoring because they improve efficiency, flexibility, and real-time control. One of the commonly used wireless communication methods in IoT applications is Bluetooth due to its low power consumption, easy implementation, and compatibility with smartphones and embedded systems.

In this experiment, Bluetooth communication was implemented using the ESP32 microcontroller, which has built-in Bluetooth capability. The ESP32 was connected to a potentiometer sensor and an MG996R servo motor to create a simple wireless monitoring and control system. The potentiometer acted as the input sensor by providing analog data to the ESP32, while the servo motor functioned as the output actuator that responded to commands sent from a smartphone through Bluetooth communication.

Task 1 focused on understanding how Bluetooth technology can be integrated into IoT applications for real-time monitoring and wireless control. The experiment involved assembling the circuit, programming the ESP32 using Arduino IDE, and establishing Bluetooth communication between the ESP32 and a smartphone using the Serial Bluetooth Terminal application. Through this experiment, the functionality of Bluetooth communication in controlling actuators and transmitting sensor data wirelessly was successfully demonstrated.

TASK 1

PART A: BLUETOOTH INTEGRATION IN IOT APPLICATIONS

Bluetooth is a wireless communication technology that allows devices to exchange data over short distances using radio frequencies in the 2.4 GHz band. In IoT applications, Bluetooth is widely used because it enables communication between smart devices such as smartphones, sensors, microcontrollers, and actuators without requiring wired connections.

In this experiment, the ESP32 acts as the central IoT device with built-in Bluetooth capability. The smartphone connects to the ESP32 through Bluetooth Serial communication using a mobile application such as Serial Bluetooth Terminal. Once paired, the smartphone can receive sensor readings from the ESP32 and also send commands back to the microcontroller. The Bluetooth communication process in IoT applications involves three main stages:

1. Data collection from sensors.

2. Wireless transmission of data through Bluetooth.
3. Remote monitoring and control using a smartphone or computer.

This method is commonly used in smart home systems, wireless monitoring systems, healthcare devices, and industrial automation because it provides low power consumption, simple implementation, and real-time communication.

PART B: INTEGRATION OF SERVO MOTOR AND POTENTIOMETER SENSOR WITH ESP32 VIA BLUETOOTH

1.0 MATERIALS AND EQUIPMENTS

- ESP32 development board
- Potentiometer sensor
- Smartphone with Serial Bluetooth Terminal Application
- Servo motor
- Breadboard
- Jumper wires

1.1 EXPERIMENTAL SETUP

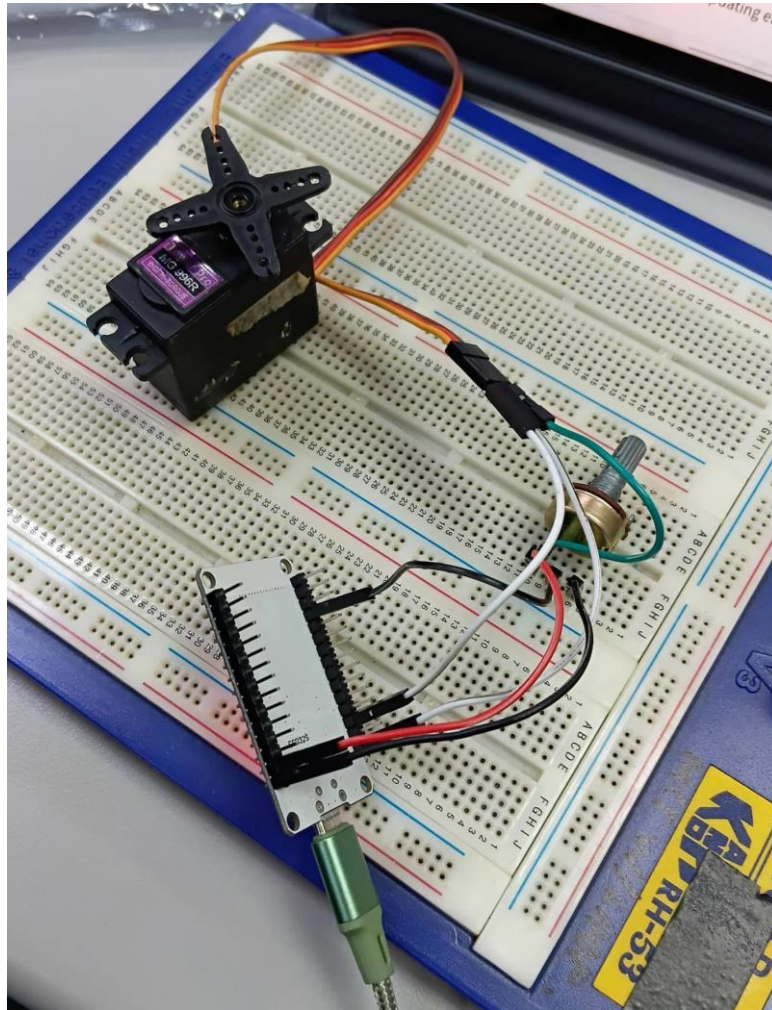


Figure 1 shows the assembly circuit for part B.

1.2 METHODOLOGY

a. Circuit Assembly

- The VCC of potentiometer is connected to 3.3 V of ESP 32
- The signal of potentiometer is connected to GPIO 34 of ESP 32
- The GND of potentiometer is connected to GND of ESP 32
- The signal of servo motor is connected to IO18 of ESP 32
- The GND of servo motor is connected to GND of ESP 32
- The VCC of servo motor is connected to 3.3 v of ESP 32

b. Programming and Uploading

- The serial Bluetooth terminal by Kai Morich from google playstore
- The serial monitor on the Arduino is being opened with desired baud rate
- The enable button of ESP 32 is being pressed
- The devices are being added to the serial Bluetooth terminal

c. Programming Logic

- The system read the potentiometer value and save in variable potValue.
- The value will be checked and then the system produces the output as:
 - If the value is less than 1365: set servo to 0°.
 - If the value is less than 2730: set servo to 90°.
 - If the value is less than 4095: set servo to 180°.
- The data will be sent to Smartphone with Serial Bluetooth Terminal Application and Serial Monitor.
- The system will repeat the process continuously.

d. Coding

```
1  #include "BluetoothSerial.h"
2  #include <ESP32Servo.h>
3
4  BluetoothSerial SerialBT;
5  Servo myServo;
6
7  const int potPin = 34; // Potentiometer input
8  const int servoPin = 13; // Servo signal pin
9  int potValue = 0;
10
11 void setup() {
12     Serial.begin(9600);
13     SerialBT.begin("Group_11"); // Change to your group name
14
15     myServo.attach(servoPin); // Attaching the servo
16     Serial.println("Bluetooth and Servo initialized.");
17 }
18
19 void loop() {
20     // Read sensor value
21     potValue = analogRead(potPin);
22
23     // Logic for Servo positions based on input ranges
24     if (potValue >= 0 && potValue < 1365) {
25         myServo.write(0);
26     }
27     else if (potValue >= 1365 && potValue < 2730) {
28         myServo.write(90);
29     }
30     else if (potValue >= 2730 && potValue <= 4095) { // Adjusted for standard 10-bit range
31         myServo.write(180);
32     }
33
34     // Send data to Smartphone via Bluetooth
35     SerialBT.print("Sensor Value: ");
36     SerialBT.println(potValue);
37
38     // Also print to local Serial Monitor for debugging
39     Serial.print("PotValue: ");
40     Serial.println(potValue);
41
42     delay(500); // Small delay for stability
43 }
44 }
```

1.3 DATA ANALYSIS

The potentiometer sensor value on serial monitor	Degree that the servo motor rotates (°)
1346 <= sensor value => 2094	The servo motor will rotate 90 °
Sensor value>2094	The servo motor will rotate to 360 ° repeatedly
Sensor value< 1346	The servo motor will rotate to initial position.

When the potentiometer is being rotated until 1346 and 2094 value is being display on serial monitor of the mobile phone, the servo motor will rotate until 90°. However, when the potentiometer is being

rotated more than 2094, the servo motor will rotate to 360° and when the potentiometer is being rotated under 1346, the servo motor will rotate to its initial position

1.4 RESULTS

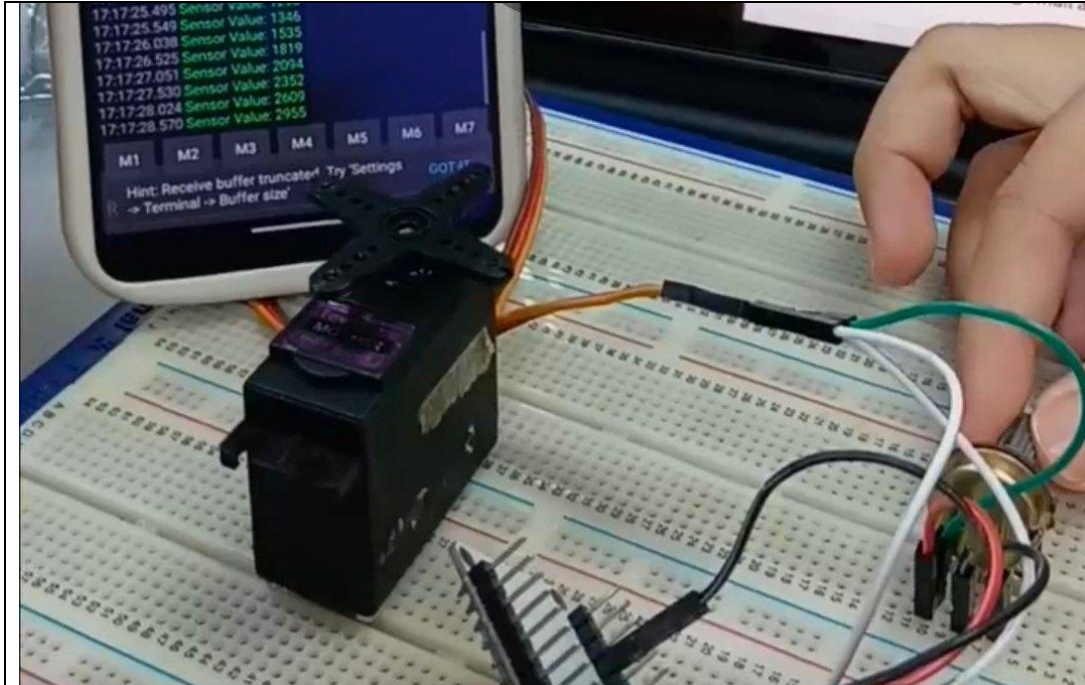


Figure 1.2 shows the serial monitor shows us the sensor value is bigger and equal with 1346



Figure 1.3 shows the serial monitor shows us the sensor value is bigger than 2094 and the servo keeps rotating 360



Figure 1.4 shows the serial monitor shows us the sensor value is 0 and the servo is rotating to its initial position

1.5 DISCUSSION

Bluetooth is usually being used in IOT applications because it will balance the low power consumption and reliable data exchange. Bluetooth Low Energy (BLE) was suitable for IOT applications. The Bluetooth can be incorporated into IOT applications by wireless sensor networks applications. For example, the smart home set up, sensors assemble the data and transfer it to a smartphone. It will experience low duty cycle which the IOT devices will spend their most time in sleep mode and will turn on when there is a burst of data. This eventually will save a lot of energy.

For other IoT Bluetooth often serves as the "onboarding" tool for other IoT protocols. For an instance, a lot of IoT devices apply Bluetooth applications to connect to the user phone first. The Bluetooth link will transmit the data to device's Wi-Fi credentials, and the Bluetooth connection will end as soon as it connects with the internet.

From the data of the task 1, when the value is less than 1346, the system believes it is in low state voltage signal, and the command will make it return to the initial position (0°). This will make sure the motor has a starting point. Next when the ($1346 \leq \text{Value} \leq 2094$) the servo will rotate to 90° , this can be applied in automated latching applications because a specific action can be achieved using desired input. Finally, when the value exceeds 2094, the servo motor will keeps rotating 360° , this will eventually transform the device from a positioning tool into a drive motor.

2.0 CONCLUSION

The integration of the ESP32 with a servo motor and Bluetooth communication successfully demonstrates a functional wireless control system, through it highlights the technical nuances of modern microcontrollers. While the logic for the potentiometer was originally designed for a standard 10-bit resolution, the ESP32 operates at a 12-bit default, creating a discrepancy in data processing. Furthermore, the project underscores the importance of power management, as the mechanical demands of the servo often exceed the current supplied by the board's internal regulators. Therefore, the system

proves that while the software framework is sound, hardware performance is strictly dependent on aligning signal resolution with physical voltage requirements.

3.0 RECOMMENDATIONS

To improve system reliability and performance, it is highly recommended to implement a common ground strategy and an external 5V power source for the servo to prevent voltage drops and potential board resets. On the software side, the `analogRead()` logic should be updated using `map()` function or by adjusting the threshold values to account for the ESP32's 0 to 4095 ADC range. Additionally, for smoother operation a small capacitor could be placed across the servo's power and ground lines to filter out electrical noise. Finally, transitioning from classic Bluetooth to Bluetooth Low Energy (BLE) would significantly optimize power consumption, making the project more suitable for long-term battery-operated applications.

4.0 REFERENCES

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6.0 STUDENTS DECLARATION

We hereby declare that, except for the places where it is acknowledged, all the work presented in this report is entirely ours. We certify that, in completing this project, we have complied with the standards of academic integrity and have not engaged in any plagiarism or unethical behaviour. Every information and support source used in this work has been appropriately referenced and acknowledged.

Certificate of Originality and Authenticity

This is to certify that we are responsible for the work submitted in this report, that the original work is our own except as specified in the references and acknowledgement, and that the original work contained herein has not been done by unspecified sources or persons. We hereby certify that this report has not been done by only one individual, and all of us have contributed to the report. The length of contribution to the reports by each individual is noted within this certificate. We also hereby certify that we have read and understand the content of the total report and no further improvement on the reports is needed from any of the individual's contributor to the report. We therefore agreed unanimously that this report shall be submitted for marking, and this final report has been verified by us.

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